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LAB TASK 03

REGN NO: FA25-BCS-017
DATA STRUCTURE LAB

Lab Task 1

Write a function which reverses the order of a doubly linked list.

CODE;

```
#include <iostream>
using namespace std;
```

```
struct Node {
    int data;
    Node* prev;
    Node* next;
};
```

```
void reverseList(Node*& head) {
    Node* current = head;
    Node* temp = NULL;

    while (current != NULL) {
        temp = current->prev;
        current->prev = current->next;
        current->next = temp;

        current = current->prev;
    }

    if (temp != NULL) {
        head = temp->prev;
    }
}
```

```
void display(Node* head) {
    while (head != NULL) {
        cout << head->data << " ";
        head = head->next;
    }
}
```

```
int main() {
    Node* head = new Node{10, NULL, NULL};
    Node* second = new Node{20, head, NULL};
}
```

```

Node* third = new Node{30, second, NULL};

head->next = second;
second->next = third;

cout << "Original List: ";
display(head);

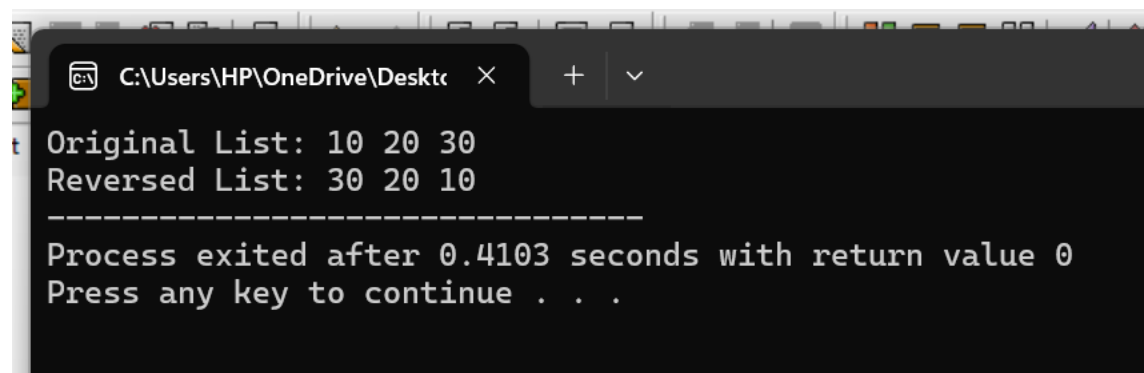
reverseList(head);

cout << "\nReversed List: ";
display(head);

return 0;
}

```

OUTPUT:



```

C:\Users\HP\OneDrive\Desktop
Original List: 10 20 30
Reversed List: 30 20 10
-----
Process exited after 0.4103 seconds with return value 0
Press any key to continue . . .

```

Lab Task 2

Write a function which takes two values as input from the user and searches them in the list. If both the values are found, your task is to swap both the nodes in which these values are found. Note, that you are not supposed to swap values.

CODE:

```

#include using namespace std;

struct Node { int data; Node* next; };

void swapNodes(Node*& head, int x, int y) {

if (x == y)
    return;

Node *prevX = NULL, *currX = head;
Node *prevY = NULL, *currY = head;

```

```

while (currX != NULL && currX->data != x) {
    prevX = currX;
    currX = currX->next;
}

while (currY != NULL && currY->data != y) {
    prevY = currY;
    currY = currY->next;
}

if (currX == NULL || currY == NULL) {
    cout << "Both values must be present!";
    return;
}

if (prevX != NULL)
    prevX->next = currY;
else
    head = currY;

if (prevY != NULL)
    prevY->next = currX;
else
    head = currX;

Node* temp = currX->next;
currX->next = currY->next;
currY->next = temp;

}

void display(Node* head) { while (head != NULL) { cout << head->data << " "; head = head->next; } }

int main() { Node* head = new Node{10, NULL}; head->next = new Node{20, NULL}; head->next-
>next = new Node{30, NULL}; head->next->next->next = new Node{40, NULL};

cout << "Original List: ";
display(head);

int x, y;

```

```

cout << "\nEnter two values: ";
cin >> x >> y;

swapNodes(head, x, y);

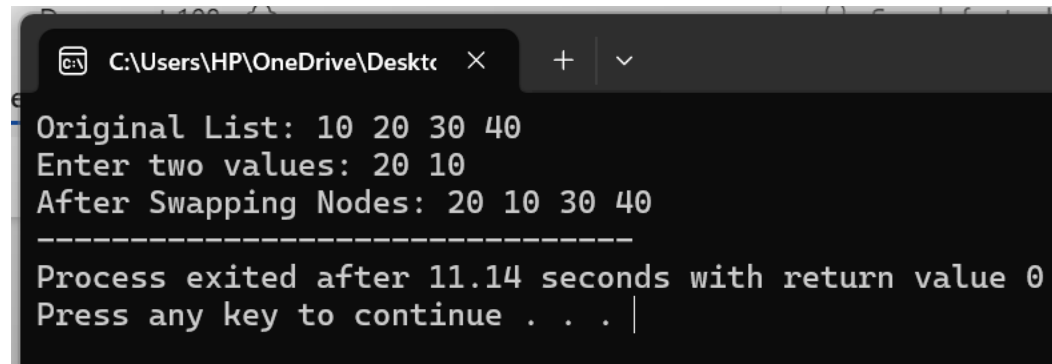
cout << "After Swapping Nodes: ";
display(head);

return 0;

}

```

OUTPUT:



```

Original List: 10 20 30 40
Enter two values: 20 10
After Swapping Nodes: 20 10 30 40
-----
Process exited after 11.14 seconds with return value 0
Press any key to continue . . . |

```

Lab Task 3

Write a function that takes a singly linked list as parameter and creates a doubly linked list for the same data present in the singly linked list i.e. for user the singly linked list will be converted into doubly linked list.

CODE:

```

#include using namespace std;

struct SNode { int data; SNode* next; };

struct DNode { int data; DNode* prev; DNode* next; };

DNode* convertToDoubly(SNode* head) {

DNode* newHead = NULL;
DNode* tail = NULL;

while (head != NULL) {

    DNode* newNode = new DNode;

```

```

        newNode->data = head->data;
        newNode->prev = tail;
        newNode->next = NULL;

        if (newHead == NULL) {
            newHead = newNode;
            tail = newNode;
        }
        else {
            tail->next = newNode;
            tail = newNode;
        }

        head = head->next;
    }

    return newHead;

}

void displayDoubly(DNode* head) { while (head != NULL) { cout << head->data << " "; head =
head->next; } }

int main() {

    // Singly Linked List
    SNode* head = new SNode{10, NULL};
    head->next = new SNode{20, NULL};
    head->next->next = new SNode{30, NULL};

    cout << "Singly Linked List: ";

    SNode* temp = head;

    while (temp != NULL) {
        cout << temp->data << " ";
        temp = temp->next;
    }
}

```

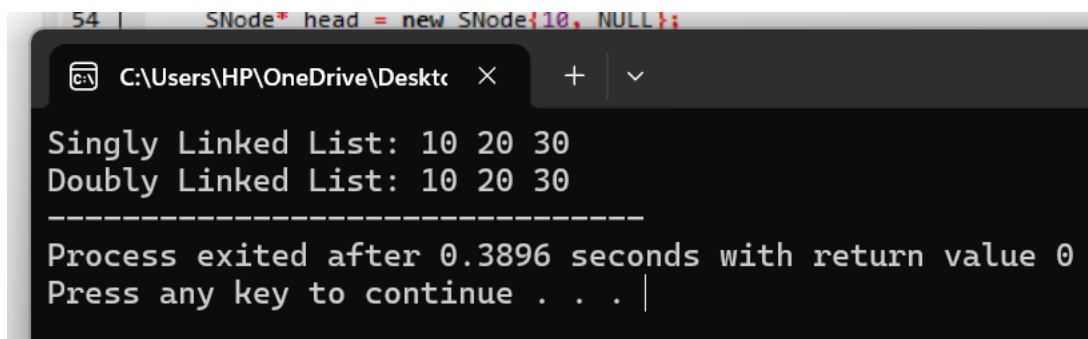
```
// Conversion
DNode* doublyHead = convertToDoubly(head);

cout << "\nDoubly Linked List: ";
displayDoubly(doublyHead);

return 0;

}
```

OUTPUT:



```
SNode* head = new SNode{10, NULL};

Singly Linked List: 10 20 30
Doubly Linked List: 10 20 30
-----
Process exited after 0.3896 seconds with return value 0
Press any key to continue . . . |
```